An Adaptive-Feedback Hénon Automorphism: Exact Dissipation and a Primitive Two-Cycle

Route-A finite certificate C122

Abstract

We construct an exact three-dimensional Hénon-type map whose additive parameter is itself a contracting feedback variable. The map has a polynomial inverse and constant Jacobian determinant 1/2. Two algebraic fixed points and one primitive period-two orbit are certified, together with its exact monodromy polynomial. Gain-zero and neighboring-gain controls show that the named orbit uses the adaptive channel. This is a finite low-period witness, not a transfer-operator or arithmetic construction.

1 Adaptive polynomial automorphism

Freeze

$$G(x, y, a) = \left(x^2 + a - y,\ x,\ \frac{a}{2} + 3x - \frac{1}{2}\right).$$

The third coordinate is an evolving parameter: it contracts and receives present-state feedback. Direct substitution gives the polynomial inverse

$$G^{-1}(X, Y, A) = \big(Y,\ Y^2 + 2A - 6Y + 1 - X,\ 2A - 6Y + 1\big).$$

The Jacobian is

$$J(x) = \begin{pmatrix} 2x & -1 & 1 \\ 1 & 0 & 0 \\ 3 & 0 & 1/2 \end{pmatrix}, \qquad \det J(x) = \frac{1}{2}.$$

Thus the model is invertible but volume contracting. These identities hold as polynomials, rather than on a finite sample alone.

2 Exact orbit witnesses

A fixed point has $y = x$, $a = 6x - 1$, and $x^2 + 4x - 1 = 0$. Hence the two certified fixed states are

$$P_\pm = (-2 \pm \sqrt{5}, -2 \pm \sqrt{5}, -13 \pm 6\sqrt{5}).$$

More importantly, the two distinct rational states

$$C_0 = (1, -1, -3), \qquad C_1 = (-1, 1, 1)$$

satisfy $G(C_0) = C_1$ and $G(C_1) = C_0$. They therefore form an oriented primitive period-two orbit. With chronological multiplication, its monodromy is

$$M = J(-1)J(1) = \begin{pmatrix} -2 & 2 & -3/2 \\ 2 & -1 & 1 \\ 15/2 & -3 & 13/4 \end{pmatrix}.$$

Consequently

$$\operatorname{tr} M = \frac{1}{4}, \qquad \det M = \frac{1}{4}, \qquad \det(I - zM) = 1 - \frac{z}{4} + \frac{5}{2}z^2 - \frac{1}{4}z^3.$$

This last polynomial is local tangent data, not a Fredholm determinant.

3 Feedback controls

Keep the two target states and parameter contraction 1/2 fixed, but write the third update as $a/2 + kx + d$. Closure at both states gives two linear equations whose unique solution is

$$k = 3, \qquad d = -\frac{1}{2}.$$

At $k = 0$ the first image has adaptive coordinate -2 rather than 1, a residual -3. At the neighboring rational gain $k = 5/2$, the residual is $-1/2$. The displayed cycle is therefore not inherited from an inert parameter coordinate.

The forward coordinate-degree prefix is

$$(2, 1, 1), (4, 2, 2), (8, 4, 4), (16, 8, 8), (32, 16, 16), (64, 32, 32),$$

which is recorded only as a finite algebraic control.

4 Validation and Route-A boundary

From the package directory run

```
python3 code/c122_adaptive_producer.py
python3 code/c122_adaptive_checker.py
python3 code/c122_sympy_crosscheck.py
python3 code/c122_replay.py
python3 code/c122_mutation.py
```

The independent checker, 21 fresh symbolic identities, canonical replay, and all 16 hostile mutations pass in exact arithmetic; eight additional symbolic boundary checks make 29 checks in total.

The canonical verdict is (`A1_WEAK`, `A2_FAIL`, `A3_FAIL`, `A4_FAIL`), overall `ROUTE_A_EXPLORATORY`. The exact low-period witnesses have no prime-like target correspondence. The tangent polynomial has no target-divisor match or analytic bridge, and no global analytic structure is established. We claim no complete orbit atlas, global transfer/Fredholm/nuclear owner, arithmetic local data, Euler factors, root numbers, automorphy, Hilbert–Pólya operator, or Route B. The firewall is `NO_BAD_EULER_OR_ROOT_NUMBER`.